AI 2002
Artificial Intelligence
Lecture 14

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FAST NUCES CFD

First Order Logic (FOL)

- The propositional logic assumes the world contains *facts* while,
- The first-order logic (like natural language) assumes the world contains,
  - *Objects*
  - *Relations*
  - *functions*.

Objects

- The **nouns and noun phrases** refer to **objects**
  - *In Wumpus-world*, the object examples are (squares, pits, and Wumpus)
  - Some other examples of objects are
    - People
    - Houses
    - Numbers
    - Theories
    - baseball
    - Games
    - centuries, etc.

Function

- **It can be unary relation such as:** red, round, is adjacent, **or n-any relation.**
- Output produce objects represented in term of theory.
- Example:

getFather(x)

It will return the name of another person.

Relation

- A relation takes terms as input and result in a sentence that wither is true or false.
- Produce Boolean output
- Example:

Brother(x,y)

First order Logic Motivation

- The statements that cannot be made in propositional logics but can be expressed with FOL.
- First-order logic can also express facts about ***some* or *all*** of the objects in the universe.

Logics

- Logics Commitment/ Language can be expressed in **two types**
    1. Ontological Commitment
    2. Epistemological Commitment

Ontological Commitment

- what exists in world
- what it assumes about the nature of reality (facts).
- Mathematically, this commitment is expressed through the nature of the model with respect to which the truth sentence is defined
- Example:
  - Propositional logic assumes that there are facts that either **hold or do not hold in the world.**
  - Suppose an AI system is designed to diagnose diseases. It has an **ontological commitment** to:
    - **Diseases exist** (e.g., diabetes, cancer).
    - **Symptoms exist** (e.g., fever, headache).
    - **Medical tests exist** (e.g., blood test, MRI).

Epistemological Commitment

- the possible states of knowledge that it allows with respect to each fact.
- In both propositional and first order logic, a sentence represents a fact and the agent either believes the sentence to be true or false, or has no opinion.
- Thus the possible values are **true/false/unknown**

Difference Ontological Commitment and Epistemological Commitment

Aspect	**Ontological Commitment**	**Epistemological Commitment**
Concerned With	What exists in reality	How knowledge is obtained
Main Question	"What things are real?"	"How do we know things?"
Example in AI	AI assumes objects like "diseases," "users," and "actions" exist	AI trusts "sensor data" or "historical records" as sources of truth
Application	Knowledge representation, logical theories	Machine learning, belief systems

Types of Logic

Different types of logics are

1. Propositional Logic
2. Temporal Logic
3. Fuzzy Logic
4. Probability Theory

Types of logic

- **Temporal Logic**
  - assumes that facts hold at particular times and
  - those times (which may be points or intervals) are ordered
  - Example: **smart traffic system** uses temporal logic to manage signals.
- **Probability Theory/Logic**
  - Systems using probability theory can have any degree of belief, ranging from 0 (total disbelief) to 1 (total belief).
- **Fuzzy Logic**
  - Fuzzy logic has a degree of truth between 0 and 1.
  - For example

1. the sentence “Vienna is a large city” might be true in our world only to a degree of 0.6 in fuzzy logic
2. A **smart thermostat** adjusts room temperature based on fuzzy rules.

Summary

Language/Logic	Ontological (What Exist in the world)	Epistemological (What an agent believes about fact)
Propositional Logic	Facts	True/false
First Order Logic	Facts, Objects, Relations	True/false/unknown
Temporal Logic	Facts, Objects, Relations, Times	True/false/unknown
Probability theory	Facts	Degree of belief [0,1]
Fuzzy Logic	Facts with the degree of truth	Known interval value

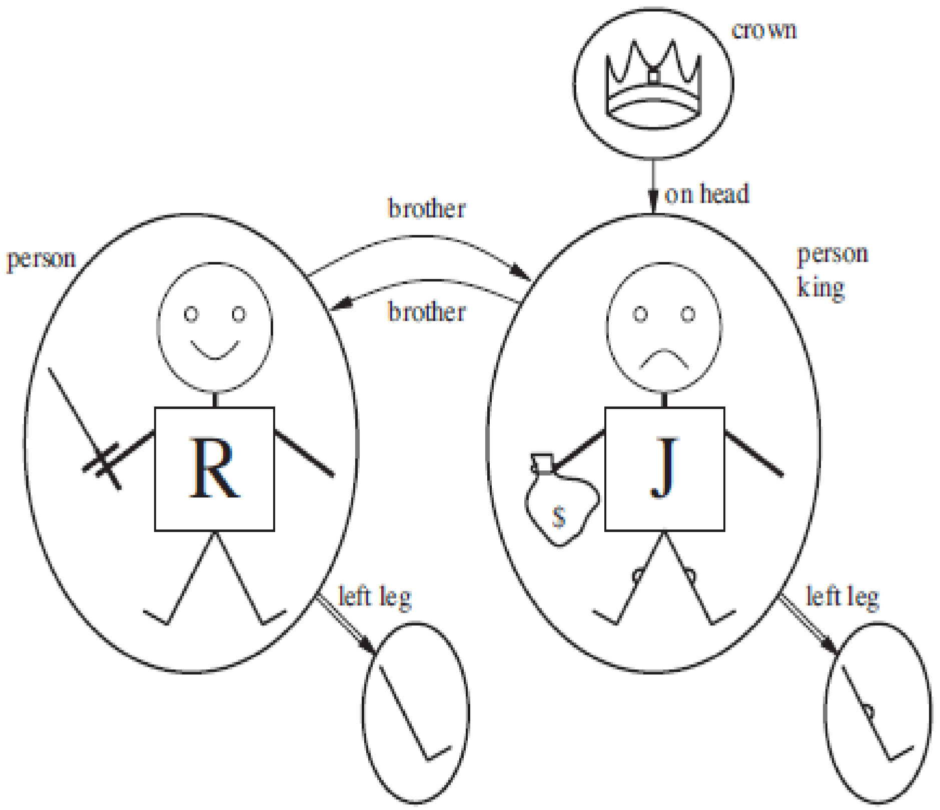
First Order Models

- Models for first-order logic have objects in them.
- If object exits in the model, so there must be **some domain of the objects**
- The domain of a model is the set of objects or domain elements it contains.
- The domain is required to be nonempty
  - every possible **world must contain at least one object.**

The **first-order logic** assumes the world contains, ***objects***, ***relations*** and ***functions***.

Models of First Order Logic

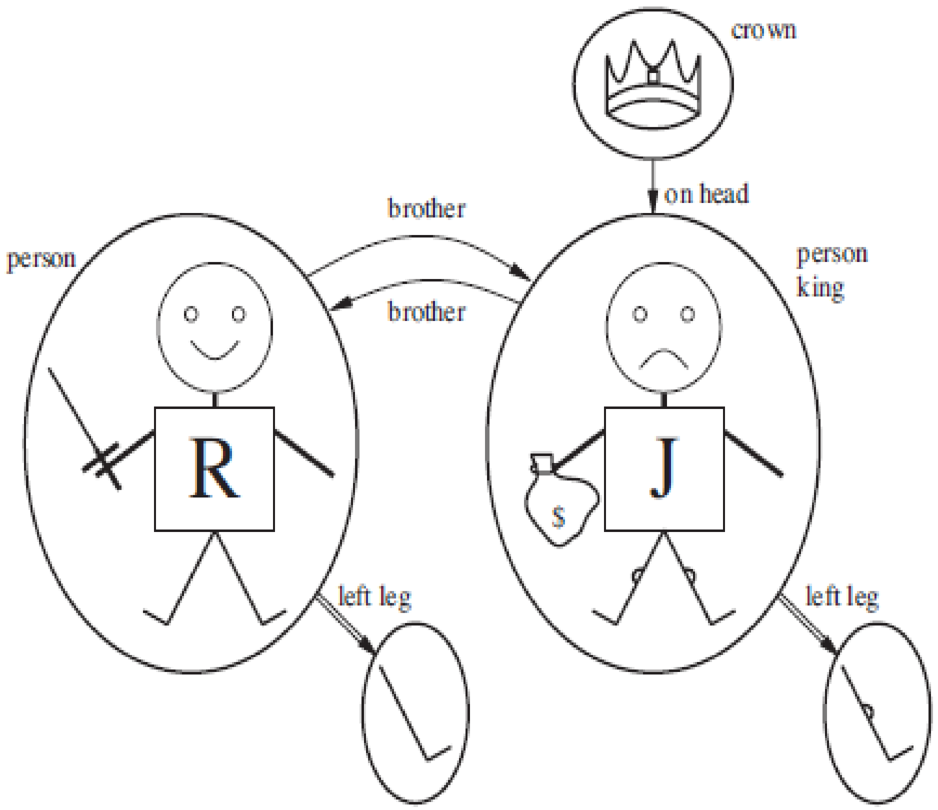
- **Richard the Loin heart** was a king of England from 1189 to 1199.
- His younger brother was the **evil king John**, who ruled from 1199 to 1215.
- The **left legs of Richard and john were different;**
- **John had a crown** because he was a king.



Example:

- The above model Contains

1. Five objects
2. Two Binary Relations = boolean
3. Three unary relations = boolean
4. One Unary function



Example

Objects : Noun and Noun Phrases

- We have 5 objects
  1. Person King John
  2. Person Richard
  3. Crown
  4. Left Leg of John
  5. Left Leg of Richard

Binary Relation : verb and Verb Phrases

We have 2 objects

1. Brother
2. On head

Unary Relations :Verb and Verb Phrases

We have 3 objects

1. Person
2. King
3. Crown

One unary function,

1. **left leg**

Example of Function and relation

Relation

- Onhead<the crown, King john>
- Brother<john, Richard>
- Crown<John>
- Person<Richard>
- King<John>

Function

- [no other person wear the crown except king]
- <John the King>- onhead(crown)

Model of First Order Logic

- Tuple
  - A tuple is the collection of objects arrange in the fixed order and is written with the angle brackets.
- Tuple Example:
  - The “brotherhood” relation in the model” is the set:

{<Richard the Lionheart, King John>, <King John, Richard the Lionheart>}.

  - The crown is on King John’s head, so the “on head” relation contains just one tuple,

<the crown, King John>.

FOL symbol and Interpretations

- Symbol
  - The basic syntactic elements of first-order logic are the symbols that stand for
  - objects, relations, and functions
  - The symbols will begin with UPPERCASE letters.
- There are three types of symbols in FOL

1. Constant Symbol
   - which stands for objects, like Richard and John
2. Predicate Symbol
   - which stands for relations, like Brother, OnHead, Person, King, and Crown
3. Function Symbol
   - which stands for relations, like Brother, OnHead, Person, King, and Crown

Syntax of FOL: Basic Elements

- Variables
  - x, y, a and b
- Connectives
  - And
  - Or
  - Not
  - Implication
  - Biconditional
- Qunatifiers
  - $\forall$, $\exists$

FOL Symbol and Interpretations

- Interpretation specifies exactly which **objects, relations, and functions** are referred to by the constant, predicate, and function symbols.

- Arity:
  - Each predicate and function symbol comes with an arity that fixes the number of arguments.

FOL Symbol and Interpretations

- Term
  - A term is a logical expression that refers to an object.

    Function(term1, term2……term n)
- A term may contain:
  - Constant symbol: Fred, Japan, Bacterium 39
  - Variables: a,b, x
  - Functional symbols are applied to one or more terms. F(x), Mother-of(John)
- A term with no variables is called a ground term.

FOL Symbol and Interpretations

- Sentence
  - A predicate symbol may be applied to terms. On(a, b), Sister(Jane, John), Sister(Mother-of(Jane), Jen)
  - $\boldsymbol{term1 \quad = \quad term2}$
  - A functional symbol may be applied to one or more terms. F(x), Mother-of(John).
  - If $\boldsymbol{v}$ is a variable and $\boldsymbol{S}$ is a sentence, then

  - $(\forall \boldsymbol{v}\ \boldsymbol{S})$ and $(\exists \boldsymbol{v}\ \boldsymbol{S})$ are sentences too.

Atomic Sentence

(or atom for short) is formed from a predicate symbol optionally followed by a parenthesized list of terms

- Syntax:

  Predicate(term1, term 2)

- Example:

  Brother(King John, Richard the Lion Heart)

- Atomic sentences can have **complex terms** as arguments.

  - **Married(Father (Richard), Mother (John))**

Atomic Sentence Example:

$$\underbrace{\underbrace{Brother}_{\text{predicate}}\,(\underbrace{\underbrace{KingJohn}_{\text{constant}}}_{\text{term}},\ \underbrace{\underbrace{RichardTheLionheart}_{\text{constant}}}_{\text{term}})}_{\text{atomic sentence}}$$

Complex Sentence

- We can use logical connectives to construct more complex sentences, with the same syntax and semantics as in propositional calculus.

- Examples:

$$\neg Brother(LeftLeg(Richard), John)$$
$$Brother(Richard, John) \wedge Brother(John, Richard)$$
$$King(Richard) \vee King(John)$$
$$\neg King(Richard) \Rightarrow King(John).$$

Quantifiers

- Quantifier in the context of AI refers to an element that expresses the quantity of items within a specified range.

- There are two types of quantifiers
  1. Universal Quantifiers (For All)
  2. Existential Quantifiers (There Exists)

Universal Quantifiers

- **“All kings are persons”** is written in first-order logic as,

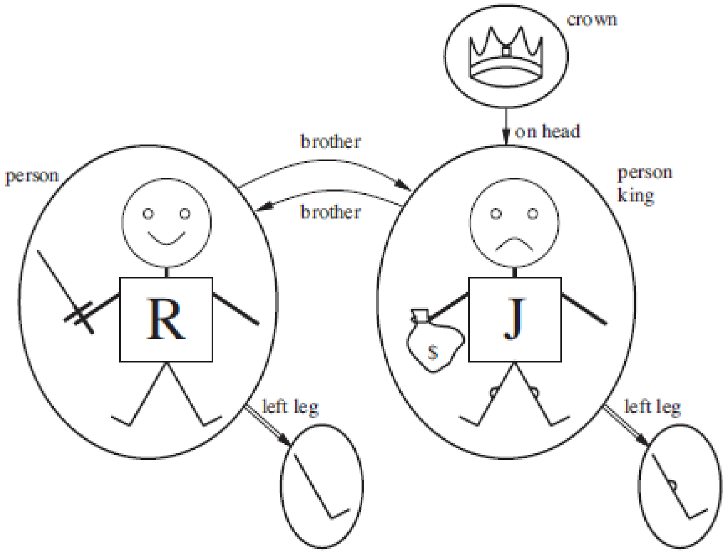
- ∀ is usually pronounced **“For all . . .”**
- *Intuitively, the sentence* **∀x P**, *where* **P** *is any logical expression, says that* **P** *is true for every object* **x.**
- More precisely,
  - **∀x P** *is true in a given model if* **P** *is true in* ***ALL*** *possible* ***extended interpretations*** *constructed from the interpretation given in the model,*
- where each extended interpretation specifies a domain element to which x refers.

Universal Quantifiers

- **“All kings are persons”** is written in first-order logic as,

- ∀ is usually pronounced **“For all . . .”**
- Example:
  - **“For all x, if x is a king, then x is a person.”**
  - We can extend the interpretation in **five** ways:

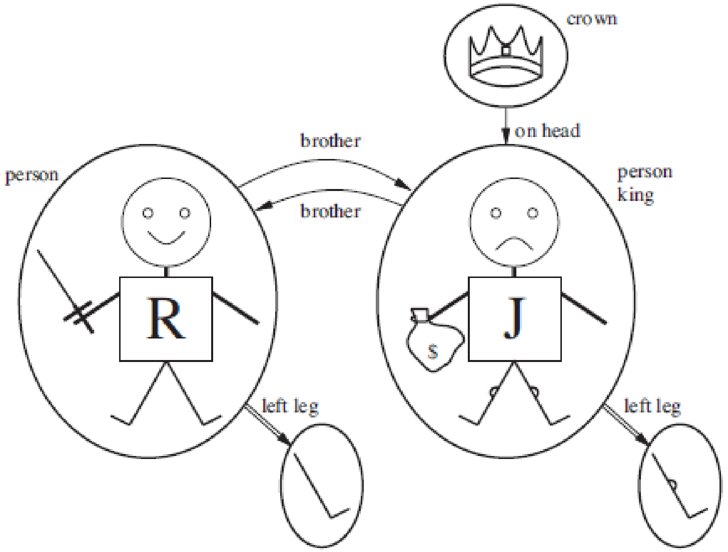
$x \rightarrow$ Richard the Lionheart,
$x \rightarrow$ King John,
$x \rightarrow$ Richard’s left leg,
$x \rightarrow$ John’s left leg,
$x \rightarrow$ the crown.



Universal Quantifiers

- The **universally quantified sentence** is equivalent to asserting the following five sentences:

Richard the Lionheart is a king $\Rightarrow$ Richard the Lionheart is a person.
King John is a king $\Rightarrow$ King John is a person.
Richard's left leg is a king $\Rightarrow$ Richard's left leg is a person.
John's left leg is a king $\Rightarrow$ John's left leg is a person.
The crown is a king $\Rightarrow$ the crown is a person.



Existential Qunatifiers

- “King John has a crown on his head”, we write

$$\exists x \;\; Crown(x) \wedge OnHead(x, John)$$

- ∃x is pronounced “There exists an x such that . . .” or “For some x . . .”.
- Intuitively, the sentence ∃x P says that P is true for at least one object x.
- More precisely, ∃ x P is true in a given true in at least one extended assigns model if P is interpretation that x to a domain element.

Existential Quantifiers

- That is, at least one of the following is true:

Richard the Lionheart is a crown ∧ Richard the Lionheart is on John's head;
King John is a crown ∧ King John is on John's head;
Richard's left leg is a crown ∧ Richard's left leg is on John's head;
John's left leg is a crown ∧ John's left leg is on John's head;
The crown is a crown ∧ the crown is on John's head.

The **fifth assertion is true in the model**, so the original existentially quantified sentence is true in the model.

Universal Quantifiers

- Typically e is the main connective with ∀.

Common Mistake:

- Using A as the main connective with ∀.

Everyone at Berkeley is smart:

$$\forall x \;\; At(x, Berkeley) \;\Rightarrow\; Smart(x)$$

$$\forall x \;\; At(x, Berkeley) \wedge Smart(x)$$

means "Everyone is at Berkeley and everyone is smart"

Existential Quantifiers

Common Mistake:

- Using e as the main connective with ∃

Someone at Stanford is smart:

$\exists x \;\; At(x, Stanford) \wedge Smart(x)$

$\exists x \;\; At(x, Stanford) \;\Rightarrow\; Smart(x)$

is true if there is anyone who is not at Stanford!

The **implication is true whenever its premise is false**—*regardless of the truth of the conclusion.*

Nested Quantifiers

- The **order of quantification is very important**. For example: **"Everybody loves somebody"** means that for every person, there is someone that person loves:

$$\forall x \; \exists y \;\; Loves(x, y)$$

- On the other hand, to say **"There is someone who is loved by everyone"** we write

$$\exists y \; \forall x \;\; Loves(x, y)$$

Nested Quantifiers

- *Some confusion may arise when two quantifiers are used with the same variable name.*
- Consider the sentence

$$\forall x \ (Crown(x) \lor (\exists x \ \ Brother(Richard, x)))$$

- Here the **x in Brother (Richard, x)** is *existentially quantified.*
- ***The rule is that the variable belongs to the innermost quantifier that mentions it; then it will not be subject to any other quantification.***

$$\exists z \ Brother(Richard, z).$$